

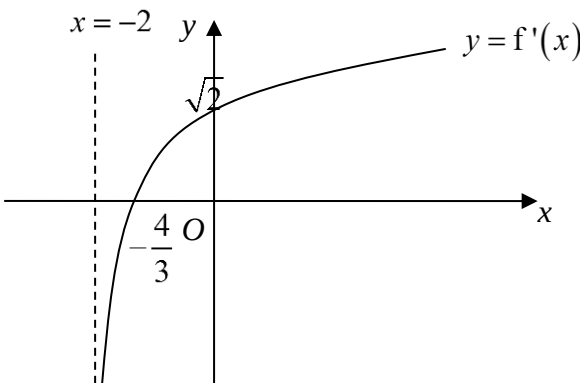
2010 'A' Level H2 Mathematics Paper 2 9740/2**Section A: Pure Mathematics [40 marks]**

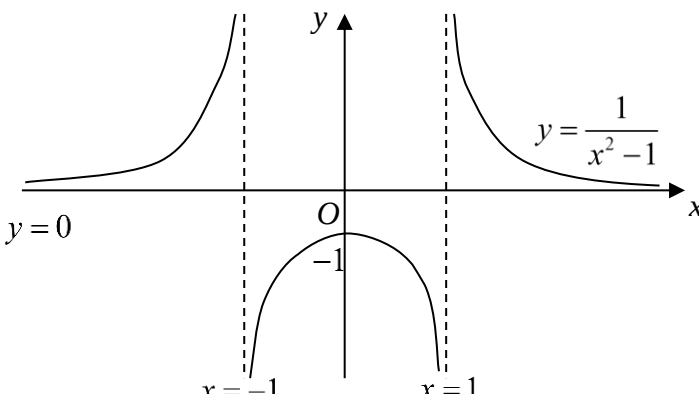
Qn 1	Suggested Solutions
(i)	Using GC, $x = 3 + 5i$ or $x = 3 - 5i$
(ii)	Let $f(x) = x^4 + 4x^3 + x^2 + ax + b$ Since $x = -2 + i$ is a root of $f(x)$, $f(-2 + i) = 0$ $(-2 + i)^4 + 4(-2 + i)^3 + (-2 + i)^2 + a(-2 + i) + b = 0$ $-12 + 16i - 2a + ai + b = 0$ Equating the real and imaginary parts: $-12 - 2a + b = 0$ $16 + a = 0$ $\Rightarrow a = -16 \quad b = -20$ $x^4 + 4x^3 + x^2 - 16x - 20 = 0$ Using GC, $x = -2 + i \text{ or } x = -2 - i \text{ or } x = 2 \text{ or } x = -2$ Use GC to evaluate $(-2 + i)^4$, $(-2 + i)^3$ and $(-2 + i)^2$. <u>Alternative Method:</u> Let $f(x) = x^4 + 4x^3 + x^2 + ax + b$ Since $x = -2 + i$ is a root of $f(x)$, and all coefficients are real numbers, $x = -2 - i$ is also a root of $f(x)$. $f(x) = x^4 + 4x^3 + x^2 + ax + b = (x + 2 - i)(x + 2 + i)(x^2 + kx + h)$ $= (x^2 + 4x + 5)(x^2 + kx + h)$ Comparing coefficients of x^3 and x^2 : $k + 4 = 4 \Rightarrow k = 0$ $h + 4k + 5 = 1 \Rightarrow h = -4$ Comparing coefficients of x and constant: $4h + 5k = a \Rightarrow a = -16$ $5h = b \Rightarrow b = -20$ Hence, $f(x) = (x^2 + 4x + 5)(x^2 - 4) = 0$ $x = -2 + i \text{ or } x = -2 - i \text{ or } x = 2 \text{ or } x = -2$

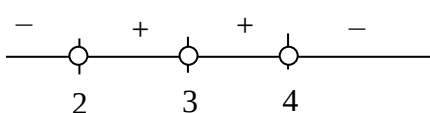
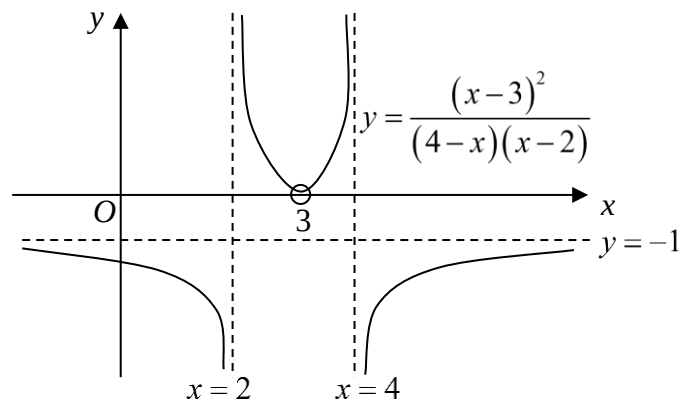
Qn 2	Suggested Solutions
(ii)(a)	$\sum_{r=1}^n \frac{1}{r(r+2)} = \frac{3}{4} - \frac{1}{2(n+1)} - \frac{1}{2(n+2)}$ Let $\frac{1}{r(r+2)} = \frac{A}{r} + \frac{B}{r+2}$ Using 'cover-up' rule, $A = \frac{1}{2}$ $B = -\frac{1}{2}$ $\therefore \frac{1}{r(r+2)} = \frac{1}{2} \left(\frac{1}{r} - \frac{1}{r+2} \right)$ $\sum_{r=1}^n \frac{1}{r(r+2)} = \sum_{r=1}^n \frac{1}{2} \left(\frac{1}{r} - \frac{1}{r+2} \right)$ $= \frac{1}{2} \left[\frac{1}{1} - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} + \frac{1}{4} - \frac{1}{6} + \dots + \frac{1}{n-2} - \frac{1}{n} + \frac{1}{n-1} - \frac{1}{n+1} + \frac{1}{n} - \frac{1}{n+2} \right]$ $= \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2} \right)$ $= \frac{3}{4} - \frac{1}{2(n+1)} - \frac{1}{2(n+2)} \text{ (proven)}$ Need to write out first 8 terms and the last 6 terms to show the cancellation.

(b)	$\sum_{r=1}^n \frac{1}{r(r+2)} = \frac{3}{4} - \frac{1}{2(n+1)} - \frac{1}{2(n+2)}$ As $n \rightarrow \infty$, $\frac{1}{2(n+1)} \rightarrow 0$, $\frac{1}{2(n+2)} \rightarrow 0$, $\sum_{r=1}^n \frac{1}{r(r+2)} \rightarrow \frac{3}{4}$ $\therefore \sum_{r=1}^{\infty} \frac{1}{r(r+2)}$ is a convergent series. $\therefore \sum_{r=1}^{\infty} \frac{1}{r(r+2)} = \frac{3}{4}$ Remember to show the reason for convergence.
------------	--

Qn 3	Suggested Solutions								
(i)	$y = x\sqrt{(x+2)}$$\frac{dy}{dx} = x \frac{1}{2\sqrt{(x+2)}} + \sqrt{(x+2)}$$= \frac{x+2x+4}{2\sqrt{(x+2)}}$$= \frac{3x+4}{2\sqrt{(x+2)}}$ You must show that it is a turning point. Solving $\frac{dy}{dx} = 0$ only gives you the stationary point(s). For the curve $y = x\sqrt{(x+2)}$ to have a turning point, $\frac{dy}{dx} = 0$$\frac{3x+4}{2\sqrt{(x+2)}} = 0$$3x+4 = 0$$x = -\frac{4}{3}$ <table><tr><td>x</td><td>$\left(-\frac{4}{3}\right)^{-}$</td><td>$\left(-\frac{4}{3}\right)$</td><td>$\left(-\frac{4}{3}\right)^{+}$</td></tr><tr><td>$\frac{dy}{dx}$</td><td>$\searrow$</td><td>$\text{—}$</td><td>$\nearrow$</td></tr></table> $\therefore$ minimum point. Since there is only 1 solution and it is a minimum point, there is only one value of x for which the curve $y = x\sqrt{(x+2)}$ has a turning point.	x	$\left(-\frac{4}{3}\right)^{-}$	$\left(-\frac{4}{3}\right)$	$\left(-\frac{4}{3}\right)^{+}$	$\frac{dy}{dx}$	$\searrow$	—	$\nearrow$
x	$\left(-\frac{4}{3}\right)^{-}$	$\left(-\frac{4}{3}\right)$	$\left(-\frac{4}{3}\right)^{+}$						
$\frac{dy}{dx}$	$\searrow$	—	$\nearrow$						
(ii)(a)	$y^2 = x^2(x+2)$$y = \pm x\sqrt{x+2}$$\frac{dy}{dx} = \pm \frac{3x+4}{2\sqrt{(x+2)}}$ At $x = 0$, $\frac{dy}{dx} = \pm \frac{4}{2\sqrt{2}} = \pm\sqrt{2}$ Use the results in (i) to identify the expression for $\frac{dy}{dx}$.								

(iii)	 Identified horizontal asymptote at $x = -2$ from $f'(x)$ expression From (i) & (ii)(a), $f'(-\frac{4}{3}) = 0$ and $f'(0) = \sqrt{2}$. For the graph of $y = f'(x)$, need to indicate the points $(-\frac{4}{3}, 0)$ and $(0, \sqrt{2})$
-------	--

Qn 4	Suggested Solutions	
(i)		
(ii)	For f^{-1} exists, f must be one-one function. Least value of $k = 0$	For f to be one-one for $x \geq k$, consider the graph when $x \geq 0$
(iii)	$fg(x) = f\left(\frac{1}{x-3}\right)$ $= \frac{1}{\left(\frac{1}{x-3}\right)^2 - 1}$ $= \frac{(x-3)^2}{1^2 - (x-3)^2}$	

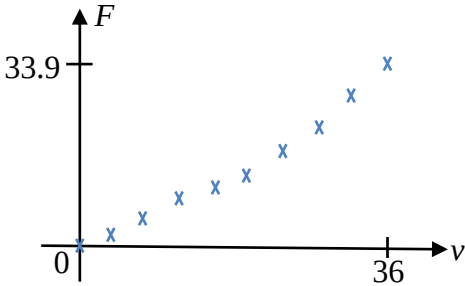
	$fg(x) = \frac{(x-3)^2}{(1+x-3)(1-x+3)}$ $= \frac{(x-3)^2}{(4-x)(x-2)} \text{ (shown)}$
(iv)	$D_{fg} = D_g = \mathbb{R} \setminus \{2, 3, 4\}$ $\therefore \frac{(x-3)^2}{(4-x)(x-2)} > 0$, for $x \in \mathbb{R}, x \neq 2, x \neq 3, x \neq 4$  $2 < x < 4, x \neq 3$ or $2 < x < 3$ or $3 < x < 4$ Alternatively, Sketch graph of $y = \frac{(x-3)^2}{(4-x)(x-2)}$  $2 < x < 4, x \neq 3$ or $2 < x < 3$ or $3 < x < 4$
(v)	$D_{fg} = D_g = (-\infty, \infty) \setminus \{2, 3, 4\} \xrightarrow{g} (-\infty, \infty) \setminus \{-1, 0, 1\} \xrightarrow{f} (-\infty, -1) \cup (0, \infty) = R_{fg}$ Alternatively, From graph, $R_{fg} = (-\infty, -1) \cup (0, \infty)$ Need to draw the graph of g to find the range of g for $x \in \mathbb{R} \setminus \{2, 3, 4\}$

Qn 7	Suggested Solutions
(i)	$P(A B') = 0.8$ $\frac{P(A \cap B')}{P(B')} = 0.8$ $P(A \cap B') = (0.8)(1 - 0.6)$ $= 0.32$
(ii)	Given $P(B) = 0.6$ and $P(A \cap B') = 0.32$ from (i) $P(A \cup B) = P(A \cap B') + P(B)$ $= 0.32 + 0.6$ $= 0.92$
(iii)	$P(B' A) = \frac{P(B' \cap A)}{P(A)}$ $= \frac{0.32}{0.7}$ $= \frac{16}{35} = 0.457 \text{ (3 s.f.)}$
(iv)	Since A and C are independent, A' and C are also independent. $P(A' \cap C) = P(A') \times P(C)$ $= (0.3)(0.5)$ $= 0.15$
(v)	$P(A' \cap B) = 0.22$ $P(A' \cap C) = 0.15$ $\therefore P(A' \cap B \cap C) \leq P(A' \cap C)$ $P(A' \cap B \cap C) = P(A' \cap C \cap B) \leq P(A' \cap C)$ $\therefore P(A' \cap B \cap C) \leq 0.15$ (When $C \subset B$, the upper bound is obtained.)

Qn 8 Suggested Solutions	
(i)	Required probability = $\frac{3 \times 4!}{5!}$ $= \frac{3}{5}$
	For Q8, we will use Probability = $\frac{\text{No. of ways for event to happen}}{\text{No. of ways for possible outcomes}}$ 3 ways fit 1 st digit ≥ 3 and 4! ways to arrange last 4 digits.
(ii)	Required probability = $\frac{3! \times 2!}{5!}$ $= \frac{1}{10}$
	2! ways to arrange last 2 even digits and 3! ways to arrange first 3 digits.
(iii)	Case 1: First digit is 3 and last digit is either 1 or 5 No of ways = $1 \times 3! \times 2 = 12$ Case 2: First digit is 5 and last digit is either 1 or 3 No of ways = $1 \times 3! \times 2 = 12$ Case 3: First digit is 4 and last digit is either 1 or 3 or 5 No of ways = $1 \times 3! \times 3 = 18$ Required probability = $\frac{12 + 12 + 18}{5!}$ $= \frac{7}{20}$

Qn 9 Suggested Solutions	
(i)	$X \sim N(180, 30^2)$ $Y \sim N(400, 60^2)$ $Y - 2X \sim N(400 - 2(180), 60^2 + 2^2(30^2))$ $\Rightarrow Y - 2X \sim N(40, 7200)$ $P(Y > 2X)$ $P(Y - 2X > 0)$ $= 0.68132$ $= 0.681 \text{ (3 s.f.)}$
	Points to note: $\text{Var}(2X) = 2^2 \text{Var}(X)$ $\text{Var}(X_1 + X_2) = 2\text{Var}(X)$ if X_1 and X_2 are independent.
(ii)	Let P be the cost of peak-rate telephone calls $P = 0.12X$ $P \sim N(0.12(180), 0.12^2(30^2))$ $\Rightarrow P \sim N(21.6, 12.96)$ Let C be the cost of peak-rate telephone calls $C = 0.05Y$ $C \sim N(0.05(400), 0.05^2(60^2))$ $\Rightarrow C \sim N(20, 9)$ Let T be the total cost of Ken's calls $T = P + C$ $T \sim N(21.6 + 20, 12.96 + 9)$ $\Rightarrow T \sim N(41.6, 21.96)$

	$P(T > 45)$ $= 0.23406$ $= 0.234$ (3 s.f)
(iii)	$P_1 + P_2 \sim N(2 \times 21.6, 2 \times 12.96)$ $\Rightarrow P_1 + P_2 \sim N(43.2, 25.92)$ $P(P_1 + P_2 > 45)$ $= 0.36183$ $= 0.362$ (3 s.f)

Qn 10	Suggested Solutions
(i)	
(ii)(a)	$r = 0.9860$
(b)	$r = 0.9907$
(iii)	$F = c + dv^2$ is the better model since r is closer to 1 as compared to $F = a + bv$. The scatter diagram also shows as v increases, F increases at an increasing rate, indicating that a linear model might not be appropriate. Scatter diagram and r justified $F = c + dv^2$ is a better model.
(iv)	A suitable regression line is F on v^2. Using GC, $F = 3.1957 + 0.024242v^2$ When $F = 26.0$, $v = 30.7$ units (3 s.f) Neither the regression line of v on F nor the regression line of v^2 on F should not be used, as v is the independent variable. Therefore, the regression line of F on v^2 should be used regardless for estimating F or v.